

Frames and matrix designs

Polynomial-time deterministic restricted isometries with nearly linear row count

Difficulty: extreme **Importance:** broadly interesting

Status: source-stated open construction problem; no resolution found in screening on 2026-09-08.

Does a deterministic algorithm exist, with running time polynomial in N , which, for every pair of integers $N \geq 2$ and $1 \leq s \leq N$, constructs a matrix $A \in \mathbb{R}^{m \times N}$ satisfying

$$m \leq Cs \log^c(2N), \quad \|x\|_2 \leq \|Ax\|_2 \leq 2\|x\|_2 \quad \text{whenever } x \in \mathbb{R}^N, |\text{supp}(x)| \leq s,$$

for absolute constants $C, c > 0$? The algorithm must output the entries, rather than merely assert existence. As usual, one may use rational output of polynomial bit length; the fixed distortion interval permits constant-accuracy rounding of a construction with slack. The row bound is permitted to exceed N , when the identity is sufficient.

This is the deterministic construction question in the data-streams workshop report, with its original fixed-distortion convention. It does not strengthen the source to the sharper $O(s \log(N/s))$ row target. Such matrices would give short, reproducible measurement operators with uniform control of every sparse least-squares subproblem.

References

1. *Open Problems in Data Streams and Related Topics*, IITK Workshop on Algorithms for Data Streams, 2006. Martin Strauss's restricted-isometry problem, first numbered construction question on printed p. 10, asks for a polynomial-time deterministic $s \text{ polylog}(N) \times N$ construction. [Workshop report](#).
2. A. S. Bandeira, *Ten Lectures and Forty-Two Open Problems in the Mathematics of Data Science*, lecture notes, Open Problem 5.1 and its surrounding discussion of explicit restricted isometries. [Author's notes](#).
3. A. S. Bandeira, M. Fickus, D. G. Mixon, and P. Wong, *The road to deterministic matrices with the restricted isometry property*, *Journal of Fourier Analysis and Applications* 19 (2013), pp. 1123–1149. Sections 1 and 4 discuss the deterministic obstruction and candidate frames. [Paper](#).

Status check — 2026-09-08

Searched “deterministic RIP polynomial time nearly optimal 2025 2026”, “restricted isometry deterministic square root bottleneck 2026”, and current explicit sensing-matrix papers. The August 2026 paper *Compressed sensing matrices from orthogonal spaces over finite fields of odd characteristic* (arXiv:2608.23062) gives particular finite-field constructions and coherence bounds, not the displayed uniform nearly linear row count. No resolution was found.